

IEEE WIE ILS 2026

ROUND 1 IDEA/SOLUTION DOCUMENT

National Hackathon

IDEATE. INNOVATE. IMPACT.

POWERING SOLUTIONS. EMPOWERING WOMEN. SHAPING TOMORROW.



IEEE WIE ILS 2026

National Hackathon

Empower to Inspire: Thriving Together for an Inclusive and Sustainable Future

Team Name Runtime riders	Track Track 1
Problem Statement / Title THE MIDDLEMAN KILLER	Solution Type: Software

What's Included
Team Information
1. Problem Understanding
2. Proposed Solution / Idea
3. Innovation & Uniqueness
4. Feasibility & Implementation Plan
5. Expected Impact
6. Technology Stack
7. Team Member Roles
Originality & Declaration

Judging Criteria (Round 1)

Criterion (Weight)	What judges look for
Problem Understanding (10%)	Clarity, evidence, relevance to inclusion/sustainability
Solution & Innovation (25%)	Originality vs. existing solutions; how directly it solves the stated problem
Feasibility & Implementation (20%)	Realistic plan, timeline, resource awareness, risk mitigation
Expected Impact (20%)	Reach, measurability, alignment with inclusion/sustainability goals
Technology & Execution (15%)	Appropriate tech/hardware choices, prototype/demo maturity
Team & Presentation (10%)	Clear roles, well-organized submission, originality declaration complete

Team Information

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Round 1 — Idea / Solution

1. Problem Understanding

The real issue isn't just that middlemen exist—it's that farmers don't have clear, transparent choices when it's time to sell. Small and scattered farmers often can't see who's buying, what the demand looks like, or what prices are actually competitive. That blind spot weakens their bargaining power and leaves them uncertain about getting a fair price. Our solution fixes this by connecting farmers with multiple verified buyers and competitive bids, so they can make selling decisions based on real information instead of guesswork. This matters for inclusion because it opens up wider markets and more buyer options for smaller farmers who'd otherwise be stuck with whoever shows up. And it matters for sustainability too, since better matching between what's being produced, what's actually in demand, and how logistics get handled makes the whole agricultural supply chain run more efficiently.

2. Proposed Solution / Idea

The Middleman Killer is a farmer-first digital platform that cuts out the guesswork by connecting farmers directly with multiple verified buyers through a transparent, competitive bidding system. Here's how it works: farmers list their produce with basic details—crop type, quantity, quality, location, and when it's available—and verified buyers then place bids on it. Instead of being stuck

with a single local buyer or having to grab the first offer that comes along, farmers can actually compare bids side by side and pick whichever works best for them. We've kept the interface simple and accessible, and we're planning local-language support and voice-assisted features so farmers with limited digital experience can use it just as easily. It's a win on both sides—buyers get direct access to available produce, and farmers get more visibility, more choice, and more control over how they sell.

Flow:

Farmer → List Produce → Verified Buyers → Competitive Bids → Compare Offers → Farmer Chooses → Digital Transaction

3. Innovation & Uniqueness

Most agricultural marketplaces out there focus on simply connecting farmers and buyers—but that's not really the hard part. The hard part is making sure the price farmers get is actually fair, and that's where we put our focus. Take e-NAM, for example—it offers digital trading and price discovery, but it works within a fairly rigid, centralized structure. Other generic agri-marketplaces, meanwhile, are mostly just listing platforms that connect buyers and sellers without doing much to create real competition or transparency around pricing. What we're doing differently is bringing all of these pieces together—multiple verified buyers, real-time competitive bidding, visibility into buyer trustworthiness, and full control resting in the farmer's hands—into one simple, easy-to-use workflow.

The key shift is this: farmers aren't handed a price and told to take it or leave it. Instead, buyers compete for their produce, and that competition is what surfaces a genuinely fair market price.

If there's one thing that makes our approach stand out, it's this combination of a farmer-controlled live auction with real competition among verified buyers—putting visibility, choice, and control back in the farmer's hands at the exact moment it matters most: the point of sale.

Single most novel element:

"We don't decide the price. We create the competition that discovers it."

Plan & Impact

4. Feasibility & Implementation Plan

We're starting with an MVP that covers the essentials: farmer and buyer registration, produce listing, verified buyer onboarding, live bidding, bid comparison, farmer acceptance, and a digital record of each transaction. Once that's working well, we'll move on to the next layer—multilingual and voice-assisted access, price intelligence, smarter buyer matching, and logistics integration.

Rather than jumping straight into large-scale deployment, we plan to start small with a pilot involving actual farmers, verified buyers, and local agricultural or FPO partners. This gives us room to learn and adjust before scaling up.

Of course, there are risks we need to stay ahead of—farmers might be slow to adopt the platform, buyer participation could be limited early on, and there's always the possibility of fraudulent users or unreliable listings. We're tackling these head-on with a design that's genuinely farmer-friendly, assisted onboarding to ease people in, thorough buyer verification, structured and clear listing information, trust scores to build confidence, and a tightly controlled pilot phase to catch issues early.

To validate the idea, we'll run it in a single region, focusing on 2–3 crop categories with a small group of users. We'll be tracking things like how many users stay active, how many auctions get created, how many bids each auction attracts, how many transactions actually go through, how long it takes to make a sale, and—just as importantly—how satisfied farmers feel with the whole experience. Once we see solid results, we'll expand into new regions and crop categories.

5. Expected Impact

The people who stand to gain the most here are small, fragmented farmers—giving them real access to verified buyers, clear price information, and competitive bids they can actually compare. Buyers benefit too, getting easier access to available produce and sellers they can trust.

For our first 12-month pilot, we're aiming to reach 500 farmers, bring on 50 verified buyers, facilitate 1,000 auction lots, and get an average of 5 or more bids per active auction. We'll be tracking success through KPIs like the number of active farmers and buyers, total auctions run, bids per auction, how many transactions actually close successfully, time to sale, how often people come back to use the platform again, and how satisfied farmers are overall.

On the inclusion side, we're lowering the barriers to entry by keeping things simple and offering multilingual and voice-assisted access—so farmers don't need to be tech-savvy to benefit. And by matching produce more accurately with actual demand and logistics, we can help agricultural goods move more efficiently through the supply chain.

Looking further ahead, we see this platform growing to cover more crops and regions, while adding features like integrated logistics, digital payments, deeper price intelligence, and stronger support systems for farmers along the way.

Technology & Team

6. Technology Stack

Software:

- **Languages:** We're building with Python, JavaScript, HTML5, and CSS3
- **Frontend:** React.js paired with Tailwind CSS and Bootstrap for a clean, responsive feel
- **Backend:** Running on Python Flask
- **Database:** PostgreSQL, managed through SQLAlchemy for the ORM layer
- **APIs & Real-Time:** REST APIs for standard communication, plus Flask-SocketIO/WebSocket to power real-time features like live bidding
- **Authentication & Security:** JWT for authentication, proper password hashing, and role-based access control to keep everything secure
- **AI/ML & Computer Vision:** Pandas and NumPy for data handling, OpenCV for image processing — with more advanced AI/ML models planned for produce price analytics and quality assessment down the line
- **Maps & Location:** OpenStreetMap and Leaflet for location-based discovery

AI/ML, IoT & Open-Source Components:

- **AI/ML:** Currently used for price analytics, with produce-quality assessment from images planned as a next step
- **OpenCV:** Handles image processing and preprocessing of produce photos before quality analysis
- **Open-Source:** OpenStreetMap and Leaflet power location-based discovery, helping farmers and buyers find each other easily

Survey / User Interviews / Field Research:

No

- We haven't conducted formal surveys or user interviews yet

- *Our current tech approach is based on the requirements and use cases we've identified so far*
- *Field research is something we plan to incorporate as we move forward*

7. Team Member Roles

Member Name	Role	Responsibilities
Patel Zeel N.	Full-Stack Developer & Project Lead	Overall project coordination; React.js frontend development; Flask backend integration; REST API development; system architecture and integration of all modules.
Patel Priyal U.	AI/ML & Computer Vision Developer	Develop price analytics; implement OpenCV-based image processing; work on AI-assisted produce-quality assessment; prepare and process datasets; integrate AI/ML models with the application.
Patel Mishal A.	Database & Backend Developer	Design and manage PostgreSQL database; develop SQLAlchemy models; implement farmer/buyer data management; authentication, JWT, role-based access control, and backend APIs.

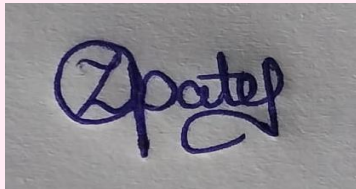
Member Name	Role	Responsibilities
Patil Krupa H.	UI/UX, Maps & Testing	Design user-friendly interfaces using Tailwind CSS/Bootstrap; integrate OpenStreetMap and Leaflet for location-based discovery; perform application testing; prepare documentation and collect user feedback.

Originality & Declaration

We, the undersigned team, declare that:

- ❖ The idea/solution submitted is original and has been conceived and developed by our team.
- ❖ This submission (or a substantially similar version) has not won a prior hackathon or competition award.
- ❖ Any third-party tools, datasets, libraries, images, or references used have been duly credited within this document.
- ❖ We understand that plagiarism, fabrication of data, or misrepresentation of authorship will lead to disqualification from IEEE WIE ILS 2026.

Team Lead Name & Signature (or e-signature):

A handwritten signature in blue ink, appearing to read "Zpater", is shown on a light-colored background.

Date: 24/08/2026
